

1. Administrative & Study Identification

Full Study Title: A Multicenter, Randomized, Parallel-group, Double-blind, Two-arm, Phase III Study to Evaluate the Safety and Efficacy of Anifrolumab Compared with Placebo in Male and Female Participants 18 to 70 Years of Age Inclusive with Systemic Sclerosis **Short Title/Acronym:** DAISY (Determine Effectiveness of Anifrolumab In SYstemic Sclerosis) **Protocol Number:** D3460C00002 **NCT Number:** NCT05925803 **Sponsor Name:** AstraZeneca **Investigational Product:** Anifrolumab (SAPHNELO) **Phase of Development:** Phase III **Medical Monitor:** Manika Azad / AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy of subcutaneous anifrolumab, compared to placebo, in reducing disease activity and skin disease in adult participants with Systemic Sclerosis (SSc). * **Secondary Objectives:** To evaluate the change in modified Rodnan skin score (mRSS), the change from baseline in forced vital capacity (FVC) in patients with SSc-associated interstitial lung disease (ILD), and to establish the long-term safety profile of anifrolumab in SSc.

2.2 Background & Rationale Systemic sclerosis is a debilitating autoimmune disease characterized by skin and internal organ fibrosis. Evidence suggests that an excess of type I interferons plays a central role in its pathogenesis and the interferon gene signature correlates with disease severity. Anifrolumab, a monoclonal antibody that targets the type I interferon receptor subunit 1 (IFNAR1), is hypothesized to reduce disease activity, skin fibrosis, and lung inflammation by blocking downstream interferon signaling in patients with SSc.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled * **Blinding:** Double-blind * **Randomization:** 1:1, Stratified by Interstitial lung disease (ILD) status at Week 0, MMF or MPA/MPS use at Week 0, and SSc disease duration (<18 months vs. $\geq$ 18 months).

3.2 Planned Enrollment & Duration * **Target N\$:** Approximately 306 evaluable patients * **Number of Sites:** Approximately 165 centers in 21 countries * **Estimated Study Start:** 08-Nov-2023 * **Estimated Primary Completion:** 06-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients from 18 to 70 years of age inclusive. 2. Diagnosed with Systemic sclerosis according to 2013 ACR/EULAR classification criteria (limited or diffuse cutaneous subsets), with disease duration within 6 years from the first non-Raynaud's phenomenon manifestation. 3. Either HAQ-DI score ≥ 0.25 points or PtGA score ≥ 3 points; AND mRSS > 10 with early disease/rapid progression OR mRSS ≥ 15 with disease duration ≥ 18 months and active disease.

4.2 Exclusion Criteria 1. Anticentromere antibody seropositivity on central laboratory testing. 2. Severe cardiopulmonary disease or history of systemic sclerosis renal crisis within the past 12 months (eGFR < 45 mL/min/1.73m²). 3. Overlap syndromes, systemic lupus erythematosus (SLE) with anti-dsDNA positivity, anti-CCP positive rheumatoid arthritis, or SSc mimics (e.g., scleromyxedema, eosinophilic fasciitis).

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** 120 mg of anifrolumab or matching placebo administered once weekly. * **Route of Administration:** Subcutaneous (SC) injection via an accessorized prefilled syringe (0.8 mL fill volume), administered at clinic or at home. * **Duration of Treatment:** 52-week double-blind, placebo-controlled period, followed by a 52-week open-label active treatment period, and a 12-week safety follow-up (116 weeks total).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Stable background standard immunosuppressant therapies, including hydroxychloroquine, mycophenolate mofetil (MMF), mycophenolic acid/sodium (MPA/MPS), methotrexate, azathioprine, tacrolimus, and low-dose oral glucocorticoids (≤ 10 mg/day). * **Prohibited:** Other investigational biologicals, targeted immunosuppressants, or unapproved concurrent autoimmune therapies not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -42 Days	Informed Consent, Medical History, classification confirmation
Baseline	Day 1 (Week 0)	Randomization, First Dose (Clinic), Baseline mRSS/FVC assessments
Interim	Weeks 1, 4, 8, 16, 24, 36, 48	Efficacy Assessments, Safety Labs, IP Dispensation. (Week 8 may be remote)
End of Study	Week 52 (Primary) & Week 116	Primary endpoint evaluation (Week 52), start of open-label phase.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * Definition: Proportion of patients achieving a Revised Composite Response Index in Systemic Sclerosis (Revised-CRISS-25) response at Week 52. *** Measurement Method:** Clinical evaluation combining mRSS, FVC, physician/patient global assessments, and HAQ-DI scores.

7.2 Secondary & Exploratory Endpoints * Change from baseline in the modified Rodnan skin score (mRSS) at Week 52. * Change from baseline in forced vital capacity (FVC) in patients with SSc-associated interstitial lung disease (ILD) at Week 52.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Method of collection via routine clinic visits, telemedicine calls, and patient reporting, actively monitoring for infections (e.g., upper respiratory infections, herpes zoster) and injection-site/infusion-related reactions.
 - **Serious Adverse Events (SAEs):** Reported by the investigator to the sponsor typically within 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to review safety data periodically throughout the study duration.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for all primary efficacy analyses; Safety Set for adverse event evaluations.
 - **Sample Size Justification:** Approximately 306 participants, randomized 1:1 to provide adequate statistical power ($\alpha = 0.05$) to detect a significant difference in Revised-CRISS-25 response rates between the anifrolumab and placebo groups.
 - **Handling of Missing Data:** Multiple Imputation (MI) and Non-Responder Imputation (NRI) rules applied for binary composite endpoints, with Last Observation Carried Forward (LOCF) for continuous variables as necessary.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

- **Monitoring Plan:** Scheduled onsite monitoring visits combined with centralized risk-based monitoring.
 - **Audit Trail:** Fully validated Electronic Data Capture (EDC) system with eCRF locking, automated query generation, and data clarification forms.
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11. Study Identifiers & Governance

- **Primary ID:** D3460C00002
 - **Registry Number:** NCT05925803
 - **Sponsor/Lead Organization:** AstraZeneca
 - **Study Title:** Determine Effectiveness of Anifrolumab In SYstemic Sclerosis (DAISY)
 - **Official Regulatory Title:** A Multicenter, Randomized, Parallel-group, Double-blind, Two-arm, Phase III Study to Evaluate the Safety and Efficacy of Anifrolumab Compared with Placebo in Male and Female Participants 18 to 70 Years of Age Inclusive with Systemic Sclerosis
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12. Clinical Framework (Systemic Sclerosis Specific)

- **Primary Condition:** Systemic Sclerosis (SSc)
 - **Diagnosis Threshold:** 2013 ACR/EULAR classification criteria for SSc
 - **Medical Methodology:** Concomitant immunosuppressant therapy (e.g., MMF, methotrexate)
 - **Optimization Target:** Blockade of the Type 1 Interferon Receptor (IFNAR1) pathway to reduce skin and lung fibrosis
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13. Study Procedures & Methodology

- **Management Pathway:** Subcutaneous (SC) biological injection pathway
 - **Education Protocol (HCP):** Site initiation and continuous training on prefilled syringe administration and standardizing mRSS assessments
 - **Education Protocol (Patient):** Injection-site training for home-based self-administration or caregiver administration
 - **Quality Audit Mechanism:** Tracking of weekly IP adherence via patient diaries and scheduled drug accountability at clinic visits
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14. Observation & Follow-Up Schedule

- **Total Duration:** 116 weeks (including 12-week safety follow-up post-treatment)
- **Onsite Visit Cadence:** Weeks 0, 1, 4, 16, 24, 36, 48, 52 (Double-Blind) followed by open-label visits through Week 104

- **Remote Monitoring Frequency:** Telephone visits permitted for interim checks (e.g., Weeks 8, 53)
- **Checklist Review Interval:** Every 8 to 12 weeks correlating with clinical visits

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				